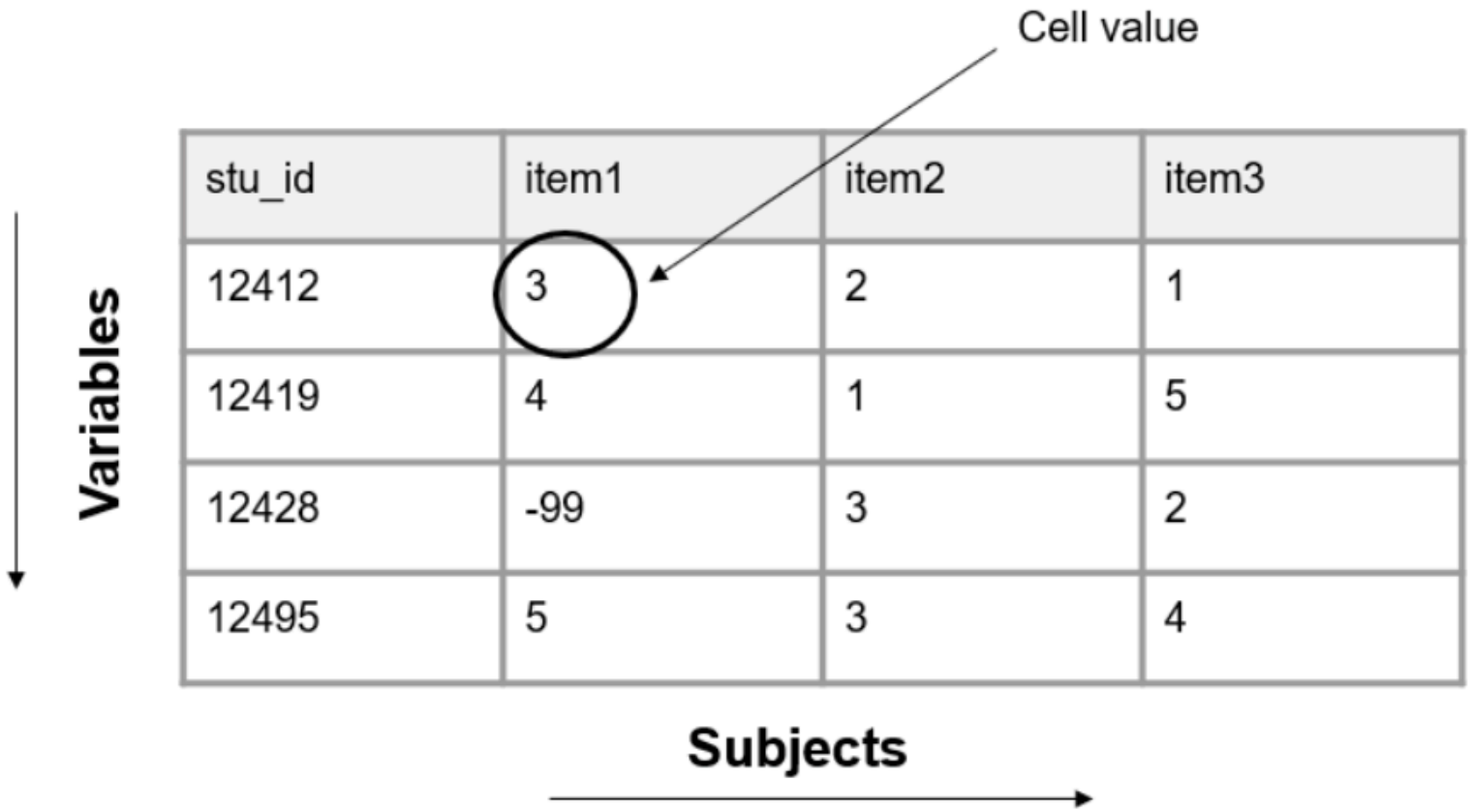
Data Management

(All content from Lewis (2024))

Rohan Alexander

21 October, 2026

Data Organization



stu_id	item1	item2	item3
12412	3	2	1
12419	4	1	5
12428	-99	3	2
12495	5	3	4

Basic format of a dataset from Lewis (2024).

Dataset organization rules

Rectangular

Not a rectangle

	12412	12419	12428	12495
age_mth	74	74	76	79
	12412	12419	12428	12495
read_raw	52	38	43	54
	12412	12419	12428	12495
read_ss	75	62	67	76

Rectangle

stu_id	age_mth	read_raw	read_ss
12412	74	52	75
12419	74	38	62
12428	76	43	67
12495	79	54	76

A comparison of non-rectangular and rectangular data from Lewis (2024).

Dataset organization rules

Column values should be consistent

Inconsistent column values

tch_id	svy_date	svy_complete
235	10-12-2023	y
236	Oct. 15, 2023	Yes
237	September 15	Y
238	2023/10/17	no

Consistent column values

tch_id	svy_date	svy_complete
235	2023-10-12	y
236	2023-10-15	y
237	2023-09-15	y
238	2023-10-17	n

A comparison of inconsistent and uniform variable values from Lewis (2024).

Dataset organization rules

Columns should adhere to your expected variable type

Character variable

tch_id	age_yr
235	22
236	24
237	49 years old
238	36..0

Space before 24 makes this entry text

Text added makes this entry text

Double decimal point makes this entry text

Numeric variable

tch_id	age_yr
235	22
236	24
237	49
238	36

A comparison of variables adhering and not adhering to a data type from Lewis (2024).

Dataset organization rules

A variable should only collect one piece of information

Two things in one variable

sch_id	level	incident_rate
31	elementary	55/250
35	elementary	72/303
42	middle	140/410
43	high	219/552

Two things in two variables

sch_id	level	incident	enrollment
31	elementary	55	250
35	elementary	72	303
42	middle	140	410
43	high	219	552

A comparison of two things being measured in one variable and two things being measured across two variables from Lewis (2024).

Dataset organization rules

All cell values should be explicit

Not explicit values

sch_id	year	grade	n_stud
31	2023	3	100
		4	80
		5	90
35	2023	3	98
		4	88
		5	91

Explicit values

sch_id	year	grade	n_stud
31	2023	3	100
31	2023	4	80
31	2023	5	90
35	2023	3	98
35	2023	4	88
35	2023	5	91

A comparison of variables with empty cells and variables with not empty cells from Lewis (2024).

Dataset organization rules

All variables should be explicit

Cell color indicates treatment condition

Not explicit variables

stu_id	date	read_raw
12412	2023-04-13	52
12419	2023-04-12	38
12428	2023-04-13	43
12495	2023-04-11	54

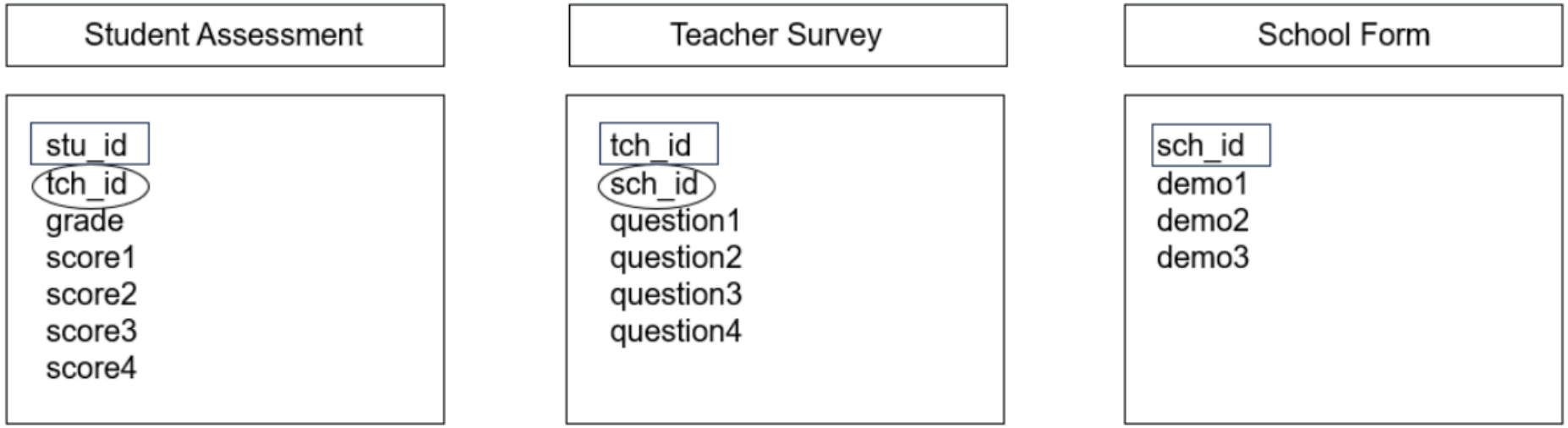
Explicit variables

stu_id	date	read_raw	treatment
12412	2023-04-13	52	0
12419	2023-04-12	38	1
12428	2023-04-13	43	1
12495	2023-04-11	54	0

A comparison of information being indicated through cell color and information being provided in an indicator variable from Lewis (2024).

Linking data

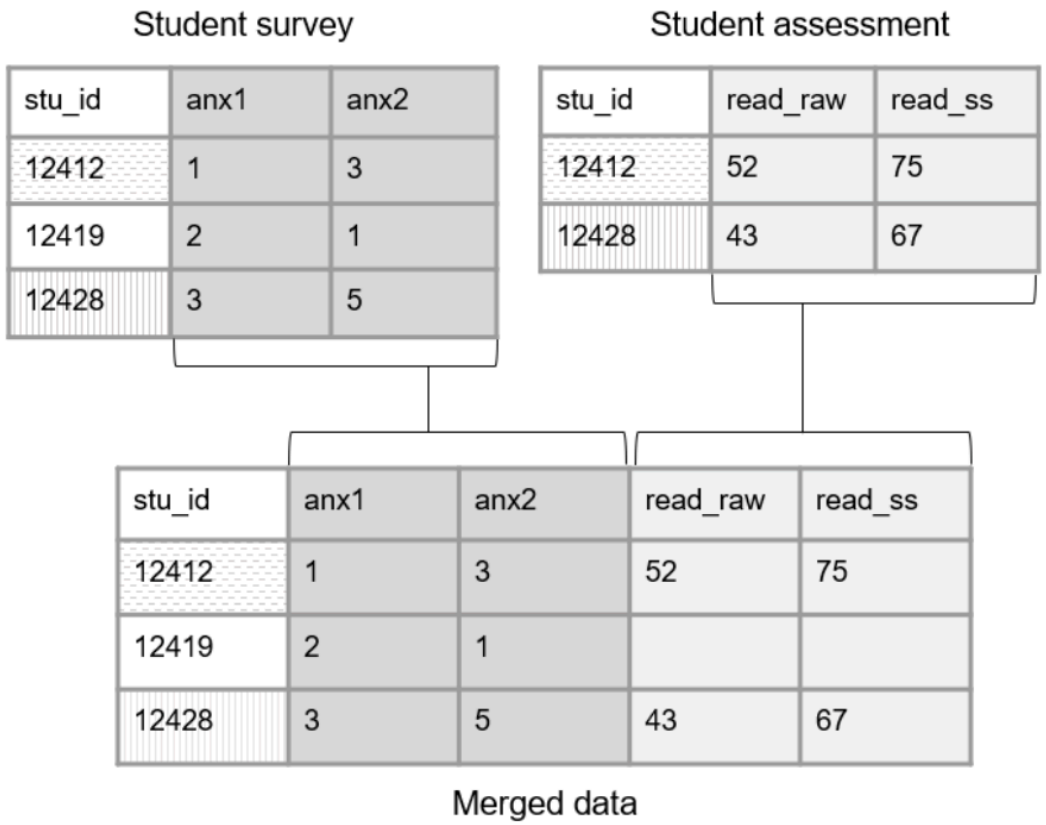
Database design



Three tables with primary and foreign keys from Lewis (2024).

Linking data

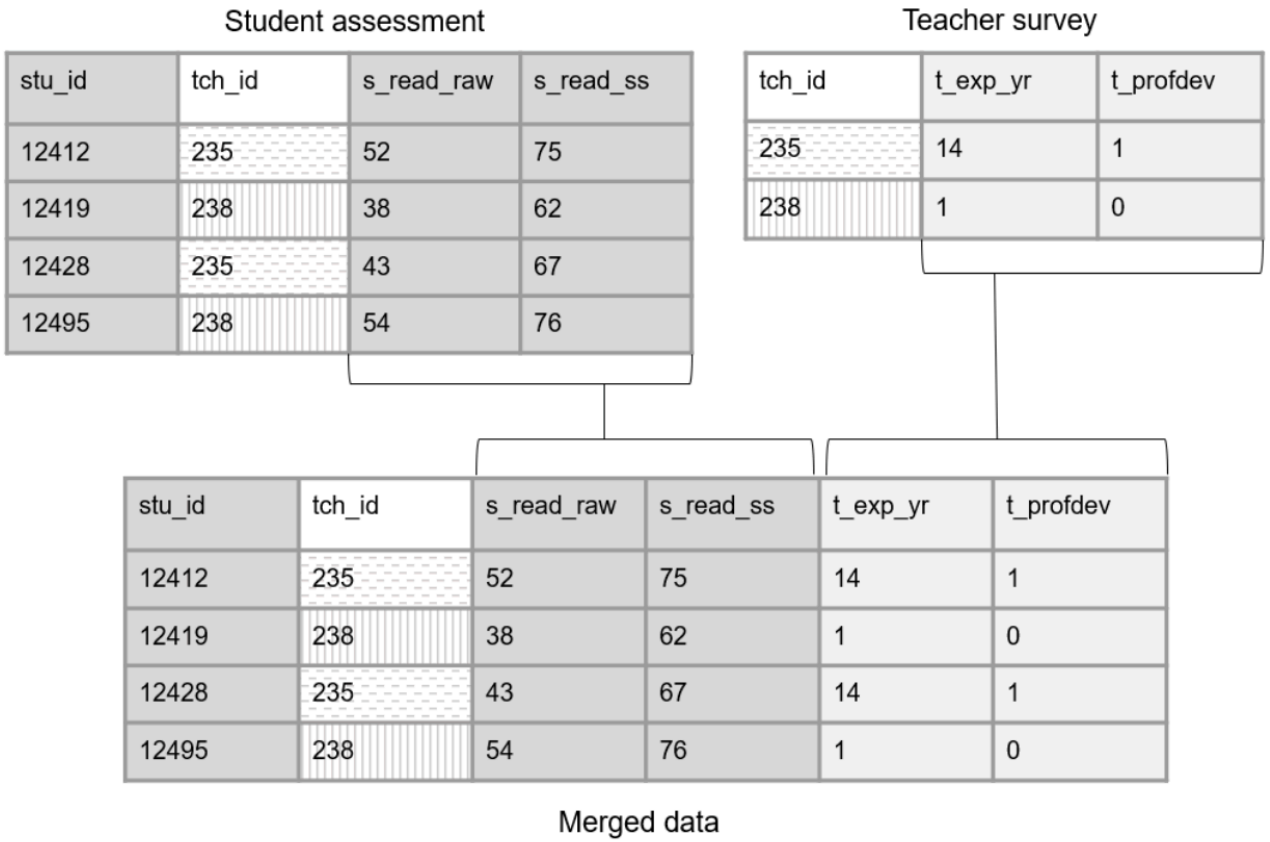
Horizontal joins



Linking data through primary keys from Lewis (2024).

Linking data

Foreign keys



Linking data through foreign keys from Lewis (2024).

Linking data

Vertical joins

Survey site 1

stu_id	sch_id	anx1	anx2
12412	31	1	3
12428	31	3	5

Survey site 2

stu_id	sch_id	anx1	anx2
12419	35	2	1
12431	35		4
12495	35	4	5

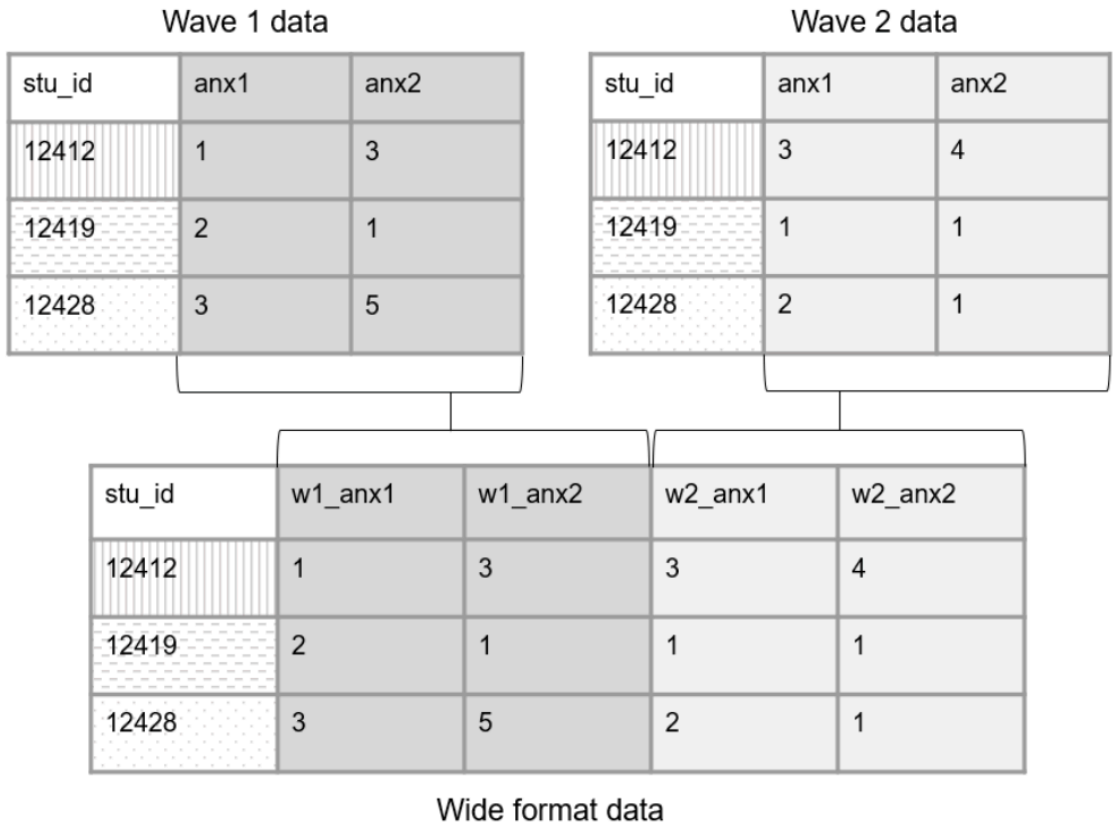
Appended data

stu_id	sch_id	anx1	anx2
12412	31	1	3
12428	31	3	5
12419	35	2	1
12431	35		4
12495	35	4	5

Appending data across sites from Lewis (2024).

Data structure

Wide format



Example linking tables across time in wide format from Lewis (2024).

Data structure

Long format

Wave 1 data

stu_id	anx1	anx2
12412	1	3
12419	2	1
12428	3	5

Wave 2 data

stu_id	anx1	anx2
12412	3	4
12419	1	1
12428	2	1

Long format data

stu_id	wave	anx1	anx2
12412	1	1	3
12419	1	2	1
12428	1	3	5
12412	2	3	4
12419	2	1	1
12428	2	2	1

Example linking tables across time in long format from Lewis (2024).

Human Subjects Data

Identifiability of a dataset

Direct Identifiers	Indirect Identifiers
Name	Age
Initials	Race
Address	Ethnicity
Phone number	Income
Email address	Education level
Social security number	Gender
IP Address	Occupation
ID numbers (student ID, state ID)	Date of birth
License numbers	ZIP Code
Account numbers	Special education services
	Data collection date
	Verbatim responses

Examples of direct and indirect identifiers from Lewis (2024).

Human subjects data oversight

- Regulations and laws
- Institutions and departments
- Agreements

Protecting human subjects data

- Get consent to collect identifiers
- Collect as few identifiers as possible
- Follow the applicable laws, policies, and agreements
- Replace names with codes
- De-identify data before sharing it
- Use data sharing agreements and controlled access

Data Management Plan

What is it?

1. Description of data to be shared
2. Format of data to be shared
3. Documentation to be shared
4. Standards
5. Data preservation
6. Access, distribution, or reuse considerations
7. Protection of privacy and confidentiality
8. Data security
9. Roles and responsibilities
10. Preregistration

Documentation

Dataset-level

1. README
2. Changelog
3. Data cleaning plan

Variable-level

1. Data dictionary
2. Codebook

Fields to Include	Optional Fields to Include
Variable name	Skip patterns
Variable label (what is this item)	Required item (were participants allowed to skip this item)
Variable type/format	Variable universe (who received this item)
Allowable values/range (including labels associated with categorical codes)	Notes (such as versions/changes to the variable)
Assigned missing values	Associated scale/subscale
Transformations	Time periods this item is available (if study is longitudinal)
Variable origin (primary/derived)	Item order
	Remove item (should this item be removed before publicly sharing data)

Fields to include in a data dictionary from Lewis (2024).

Style Guide

Style Guide

- A style guide provides general rules for the formatting of information.

General good practices

1. Avoid spaces.
2. With the exception of (_) and (-), avoid special characters.
3. Pick a naming convention.
4. Character length matters.

Directory structure

Top level content folders

```
project_new/
├── documentation
|   ├── phase1
|   |   └── ...
|   ├── phase2
|   |   └── ...
|   └── ...
├── planning
|   ├── phase1
|   |   └── ...
|   ├── phase2
|   |   └── ...
|   └── ...
└── ...
```

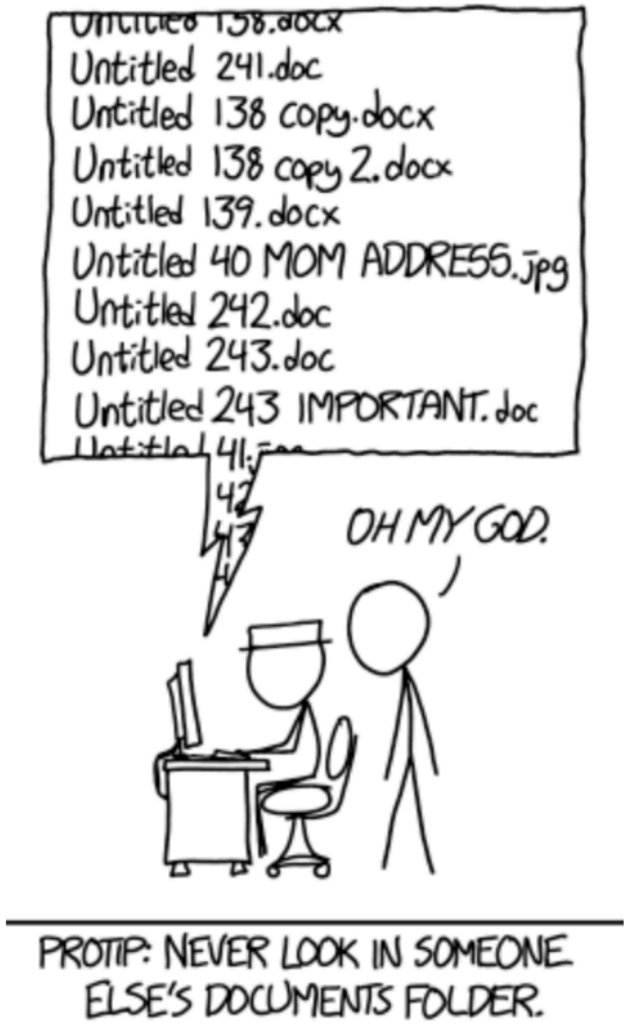
Top level content folders from Lewis (2024).

Naming folders

```
project_new/
├── 01_planning
|   ├── grant_mgmt
|   |   ├── proposal_docs
|   |   |   └── ...
|   ├── meetings
|   |   ├── agendas
|   |   |   └── ...
|   └── ...
├── 02_documentation
|   ├── consent_forms
|   |   └── ...
|   ├── data_dictionaries
|   |   └── ...
|   ├── data_sources_catalog.xlsx
|   ├── instrument_copies
|   |   └── ...
|   ├── research_protocol.docx
|   ├── sops
|   |   └── ...
|   └── ...
├── 03_data_collection
|   ├── hiring_materials
|   |   └── ...
|   ├── scheduling_materials
|   |   └── ...
|   └── ...
├── 04_tracking
|   ├── parent_consents
|   |   └── ...
|   └── participant_tracking_database.accdb
├── 05_data
|   ├── cohort1
|   |   ├── student
|   |   |   ├── survey
|   |   |   |   ├── clean
|   |   |   |   |   ├── archive
|   |   |   |   |   |   └── changelog.txt
|   |   |   |   ├── raw
|   |   |   |   |   ├── archive
|   |   |   |   |   |   └── changelog.txt
|   |   |   |   |   └── ...
|   |   |   |   └── ...
|   |   |   └── ...
|   |   └── ...
|   └── ...
└── ...
```

Top level content folders from Lewis (2024).

File naming



"Documents", from xkcd.com via Lewis (2024).

Variable naming

- Don’t use spaces or special characters, except (_).

References

Lewis, Crystal. 2024. *Data Management in Large-Scale Education Research*. 1st ed. Chapman Hall/CRC. https://datamgmtinedresearch.com/index.html.